

Sec-05

CSE360 Assignment

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Ans to the Q.No-1

(a) Tri-state is a state where only one device can access the bus at a time. If a circuit gets achieve tri-state then there will be no bus conflict. The basic concept of tri-state which is high impedance which removes the device's influence from rest of the circuit.

(b) The conflict arises because of bus conflict. In Bus Interfacing, only one I/O device can share a bus at a time as output. If more than one I/O use the bus at a time then it causes bus conflict. There are 4 causes of Bus Error and those are,

- (i) multiple device use the bus as output.
- (ii) wait states are not maintained.
- (iv) I/O device causes bus pin to stuck.
- (v) caused by glitch.

(c) USB Type-c is mostly used because it is a ~~reverse~~ reversable design which enables to connect the ~~port~~ port either way up or down. This is not available in Type-A and Type-B. It also supports faster data transfer & rate. ~~It supports~~

(d)

Serial Transmission	Parallel Transmission
1. One bit can send at a time.	1. Multiple bits can send at a time.
2. Data transfer speed is lower.	2. Data transfer speed is faster higher.
3. Data sends through a single channel.	3. Data sends to through multiple channels.

Ans to the Q. No - 2

(a) The name of the second protocol is SPI protocol.

SPI protocol supports single ~~many~~ master ~~and~~ with multiple slaves. The ~~the~~ master is actually a microcontroller. The master and slaves have ~~to~~ ~~4~~ 4 pins.

Steps,

- (i) Master first generate a clock pulse using CLK pin.
- (ii) Then Master sends a signal to a particular slave to ~~start~~ make the slave active using SS pin.
- ~~(iii) Master then sends the data to the slaves~~
- (iii) If the Master wants to send data, it uses the MOSI pin to send or write data to the slaves ~~MSI~~ MISO pin.
- (iv) If the Master wants to read ~~to~~ a data from slave then it ~~rece~~ receive the data using MISO pin.

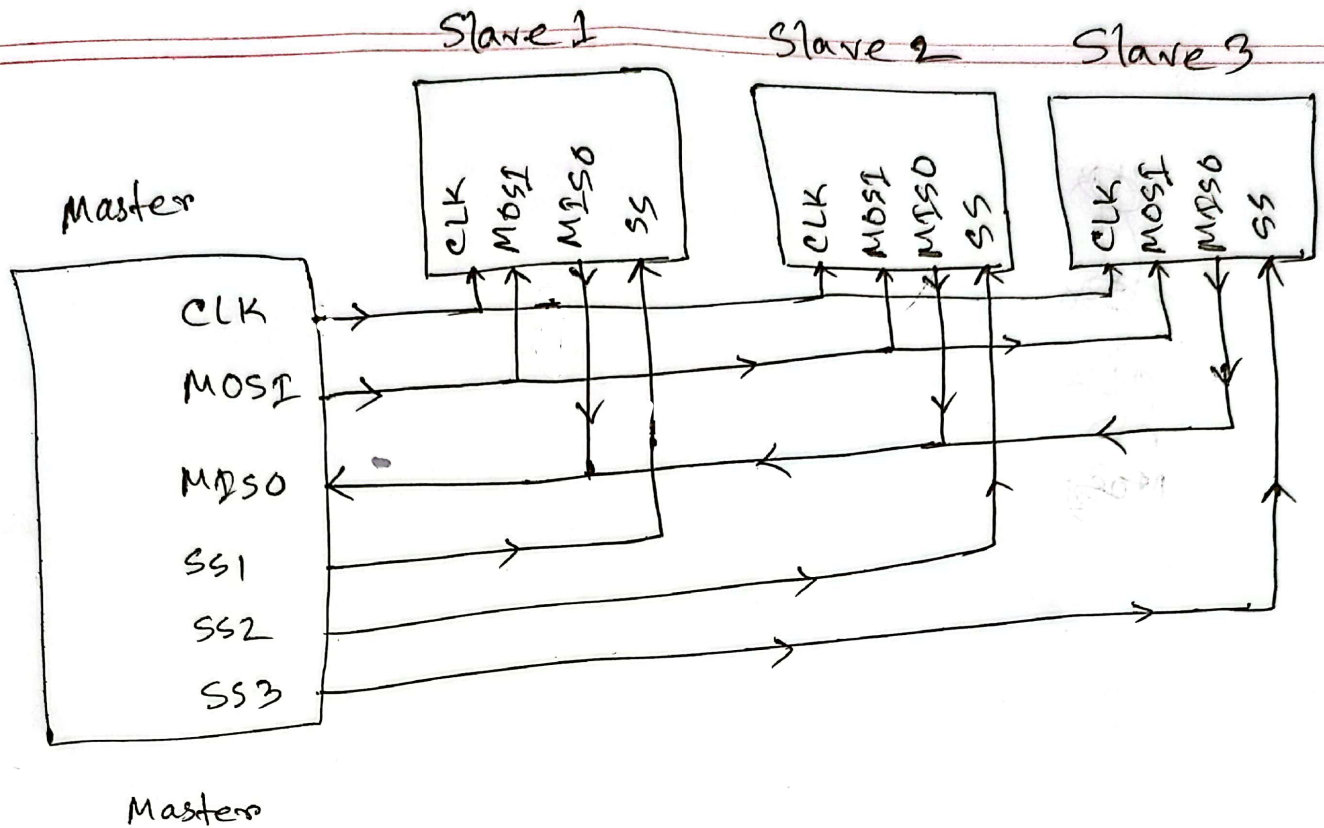


fig: SPI protocol diagram.

(b) If I want to send those three binary packets then I have to ~~select the Slave~~ ~~or~~ give a clock signal and select the Slave. then ~~now~~ send the data using MOSI pin from the Master to that Slave. At that ~~now~~ moment the data packet will look like below,

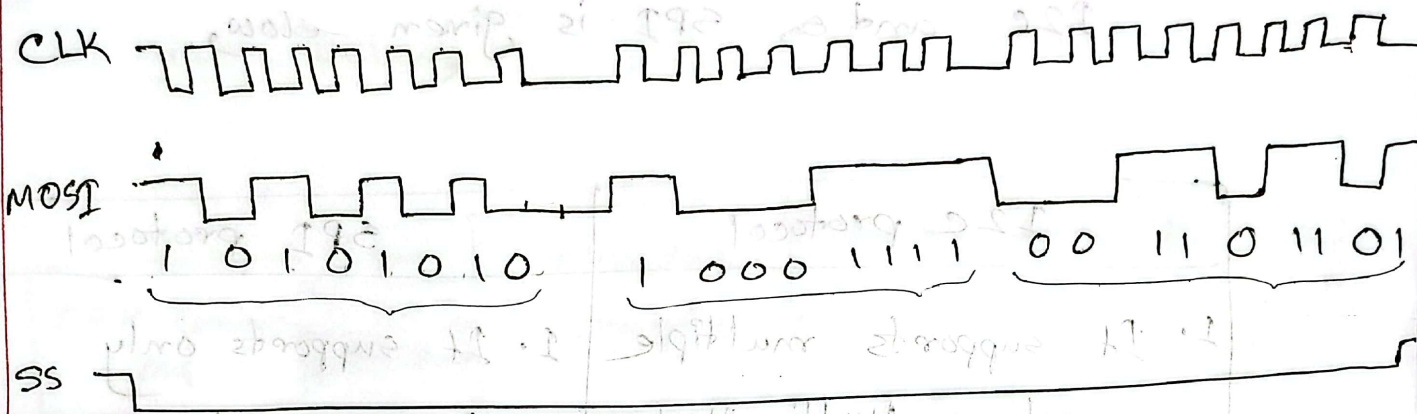


fig: timing diagram of SPI

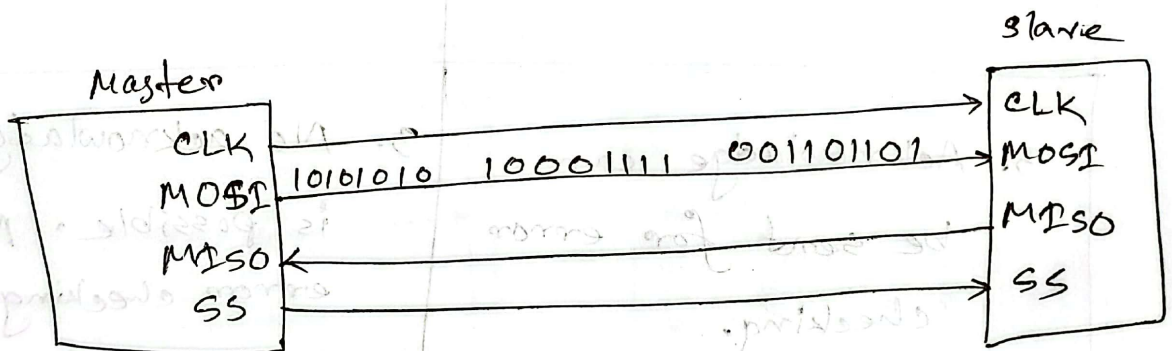


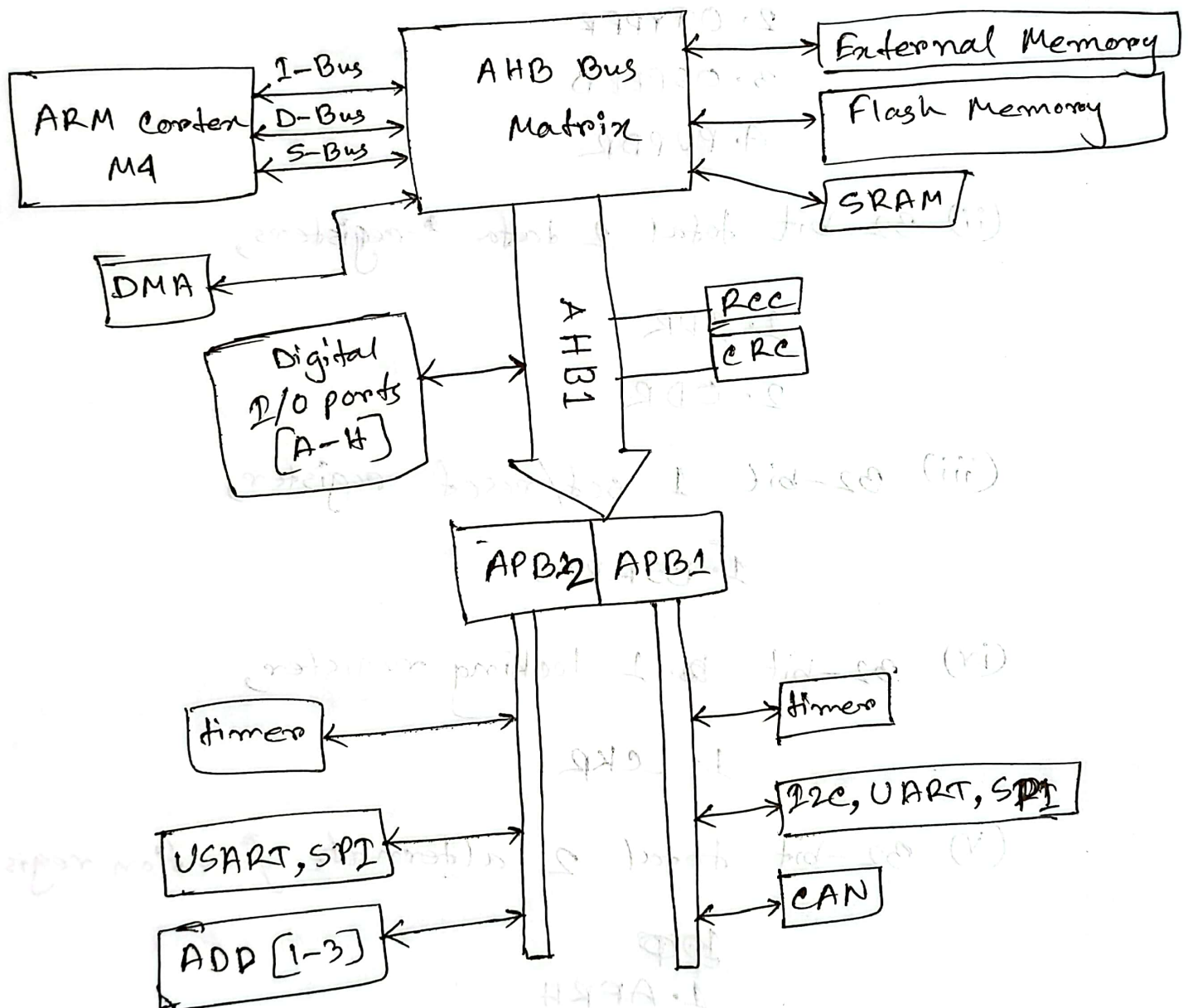
fig: packet sending through SPI

(c) The first protocol is I2C and second protocol is SPI. The comparison between I2C and ~~SP~~ SPI is given below,

I2C protocol	SPI protocol
1. It supports multiple masters. Multi-Master configuration	1. It supports only single Master.
2. The communication speed is slower.	2. The communication speed is faster.
3. Acknowledge can be sent for error checking.	3. No acknowledge ^{is} is possible. No error checking.

Ans to the Q. No-3

(a) Simplified system architecture of STM32 is given below,



(b) In STM32, there are many registers are required for GPIO operations.

(i) 32-bit total 4 configure registers,

1. MODER

2. OTYPER

3. OSPEED

4. PUPDR

(ii) 32-bit total 2 data registers,

1. IDR

2. ODR

(iii) 32-bit 1 set/reset register,

1. BSRR

(iv) 32-bit 1 locking register,

1. LCKR

(v) 32-bit total 2 alternate function registers,

~~1. AFRH~~

1. AFRH

2. AFRL

(c) The commands are written down below,

(i) To active port c: $RCC \rightarrow AHB1ENR |= (1 \ll 2)$

(ii) pin-7 for high ~~data~~ speed data output:

$GPIOC \rightarrow MODER |= (1 \ll 14)$

$GPIOC \rightarrow OTYPER |= (1 \ll 7)$

$GPIOC \rightarrow OSPEEDR |= (3 \ll 14)$

(iii) pin-9 for pull-up input:

$GPIOC \rightarrow MODER \&= \sim(3 \ll 18)$

$GPIOC \rightarrow PUPDR |= (1 \ll 18)$

(d) configuration:

(i) $RCC \rightarrow AHB1ENR |= (1 \ll 3)$: Port D is Activated

(ii) $RCC \rightarrow AHB1ENR \&= \sim(0 \ll 5)$: No effect on port activation.

(iii) $RCC \rightarrow AHB1ENR |= (1 \ll 1) |= (1 \ll 0)$: Both Port A and B is activated.

So, Port A, B and D will be activated.